

System Test Plan

Instructions. In this section, you must provide your system test plan with at least 5 test cases. **If you want to provide more than 5 test cases, add an appendix at the end of this document with the 6th, 7th, etc. test cases so that page numbers for all sections match the required template! Only the first 5 test cases will be graded.**

Make sure:

- You provide your sample test data
- Test IDs are uniquely identified and descriptive
- Test descriptions are fully specified with complete inputs, specific values, and preconditions
 - Be sure to provide SPECIFIC INPUTs and VALUEs so that your test cases are repeatable
- Expected results are fully specified with specific output values
- All tests cover scenarios based on the problem statement
- All tests cover unique scenarios for the system
- All strategies for system testing are demonstrated in the tests (testing equivalence classes, testing boundary values, testing exceptions/unexpected inputs)

Test Data:

For the system test cases, we will use the following test files named landmarkInfo.txt:

LANDMARK_ID,DESCRIPTION,TYPE

L01,Park Entrance,Location

L02,Entrance Fountain,Fountain

L03,Waste Station 1,Pet Waste Station

L04,Entrance Restrooms,Restroom

L05,Overlook 1,Overlook

L06,Rock Formation 1,Rock Formation
L07,Overlook 2,Overlook
L08,Overlook Restrooms,Restroom
L09,Waste Station 2,Pet Waste Station
L10,Hidden Gardens,Gardens
L11,Campsite 1,Campsite
L12,Campsite Restrooms,Restroom

and trails.txt:

LANDMARK_ID,LANDMARK_ID,DISTANCE
L01,L02,3013
L01,L03,1046
L01,L04,1179
L02,L10,3613
L03,L05,4204
L04,L09,2311
L05,L06,1039
L06,L07,2912
L07,L08,1891
L11,L12,1066

Test ID	Description	Expected Results	Actual Results
Test #1 testID: testGetIntersectingTrails Strategy: (Boundary value – listing trails with 1 intersection) Author: Jayani Sivakumar	Preconditions: - TrailManagerUI has been loaded successfully. - The user has loaded landmarkInfo.txt and trails.txt successfully. Steps: - Enter the number 2 in the section “Enter your answer:” that represents “View a report of the proposed landmarks at which first aid stations should be located.” and press the Enter key. - Enter 1 in the section “Enter the minimum number of intersecting trails:” and press the Enter key. Check Results	The system displays: Proposed Locations for First Aid Stations { - Park Entrance (L01) - 3 intersecting trails - Entrance Fountain (L02) - 2 intersecting trails - Entrance Restrooms (L04) - 2 intersecting trails - Overlook 1 (L05) - 2 intersecting trails - Overlook 2 (L07) - 2 intersecting trails - Rock Formation 1 (L06) - 2 intersecting trails - Waste Station 1 (L03) - 2 intersecting trails - Campsite 1 (L11) - 1 intersecting trails - Campsite Restrooms (L12) - 1 intersecting trails - Hidden Gardens (L10) - 1 intersecting trails - Overlook Restrooms (L08) - 1 intersecting trails - Waste Station 2 (L09) - 1 intersecting trails	Output: Upon entering 2 → Enter (to select “View a report of the proposed landmarks at which first aid stations should be located”) and then 1 → Enter (minimum intersections), the console printed exactly: Proposed Locations for First Aid Stations { - Park Entrance (L01) - 3 intersecting trails - Entrance Fountain (L02) - 2 intersecting trails - Entrance Restrooms (L04) - 2 intersecting trails - Overlook 1 (L05) - 2 intersecting trails - Overlook 2 (L07) - 2 intersecting trails - Rock Formation 1 (L06) - 2 intersecting trails

		} Again, the menu is re-displayed: Menu: 1. View distances to all reachable landmarks. 2. View a report of the proposed landmarks at which first aid stations should be located. 3. Exit TrailManager. Enter your answer:	• Waste Station 1 (L03) - 2 intersecting trails • Campsite 1 (L11) - 1 intersecting trails • Campsite Restrooms (L12) - 1 intersecting trails • Hidden Gardens (L10) - 1 intersecting trails • Overlook Restrooms (L08) - 1 intersecting trails • Waste Station 2 (L09) - 1 intersecting trails } Again, the menu is re-displayed: Menu: 1. View distances to all reachable landmarks. 2. View a report of the proposed landmarks at which first aid stations should be located. 3. Exit TrailManager. Enter your answer:
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Test #2 testID: testGetIntersectingTrailsInvalidNumber Strategy: (Exception/unexpected input – trying to view an invalid number of intersections) Author: Jayani Sivakumar	Preconditions: - TrailManagerUI has been loaded successfully. - The user has loaded landmarkInfo.txt and trails.txt successfully. Steps: - Enter the number 2 in the section “Enter your answer:” that represents “View a report of the proposed landmarks at which first aid stations should be located.” and press the Enter key. - Enter 0 in the section “Enter the minimum number of intersecting trails:” and press the Enter key. Check Results	“Number of intersecting trails must be greater than 0.” Again, the menu is re-displayed: Menu: - View distances to all reachable landmarks. - View a report of the proposed landmarks at which first aid stations should be located. - Exit TrailManager. Enter your answer:	Output: After choosing option 2 and inputting 0 → Enter, the system outputs: “Number of intersecting trails must be greater than 0.” Again, the menu is re-displayed: Menu: - View distances to all reachable landmarks. - View a report of the proposed landmarks at which first aid stations should be located. - Exit TrailManager. Enter your answer:
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Test #3 testID: testGetIntersectingTrailsNoTrails Strategy: (Exception/unexpected input – choosing a number of intersections with no trails) Author: Jayani Sivakumar	Preconditions: - TrailManagerUI has been loaded successfully. - The user has loaded landmarkInfo.txt and trails.txt successfully. Steps: - Enter the number 2 in the section “Enter your answer:” that represents “View a report of the proposed landmarks at which first aid stations should be located.” and press the Enter key. - Enter 4 in the section “Enter the minimum number of intersecting trails:” and press the Enter key. Check Results	“No landmarks have at least 4 intersecting trails.” Again, the menu is re-displayed: Menu: - View distances to all reachable landmarks. - View a report of the proposed landmarks at which first aid stations should be located. - Exit TrailManager. Enter your answer:	Output: Upon selecting option 2 then entering 4 → Enter, the console displays: “No landmarks have at least 4 intersecting trails.” Again, the menu is re-displayed: Menu: - View distances to all reachable landmarks. - View a report of the proposed landmarks at which first aid stations should be located. - Exit TrailManager. Enter your answer:

Test #4 testID: testViewDistancesValid Strategy: (Equivalence class – viewing distances	Preconditions: - TrailManagerUI has been loaded successfully. - The user has loaded landmarkInfo.txt and trails.txt successfully. Steps: - Enter the number 1 in the section “Enter your answer:” that represents “View distances to all reachable landmarks” and press the Enter key. - Enter “L02” as the starting landmark ID and press the	The system displays: Landmarks Reachable from Entrance Fountain (L02) { 3013 feet to Park Entrance (L01) 3613 feet to Hidden Gardens (L10) 4059 feet to Waste Station 1 (L03) 4192 feet to Entrance Restrooms (L04) 6503 feet (1.23 miles) to Waste Station 2 (L09) 8263 feet (1.56 miles) to	Output: After entering 1 and then L02 → Enter, output appears exactly as: Landmarks Reachable from Entrance Fountain (L02) { 3013 feet to Park Entrance (L01) 3613 feet to Hidden Gardens (L10) 4059 feet to Waste Station 1 (L03) 4192 feet to Entrance Restrooms (L04)

from a valid landmark ID with multiple reachable landmarks) Author: Jayani Sivakumar	Enter key. Check Results	Overlook 1 (L05) • 9302 feet (1.76 miles) to Rock Formation 1 (L06) • 12214 feet (2.31 miles) to Overlook 2 (L07) • 14105 feet (2.67 miles) to Overlook Restrooms (L08) } Again, the menu is re-displayed: Menu: 1. View distances to all reachable landmarks. 2. View a report of the proposed landmarks at which first aid stations should be located. 3. Exit TrailManager. Enter your answer:	• 6503 feet (1.23 miles) to Waste Station 2 (L09) • 8263 feet (1.56 miles) to Overlook 1 (L05) • 9302 feet (1.76 miles) to Rock Formation 1 (L06) • 12214 feet (2.31 miles) to Overlook 2 (L07) • 14105 feet (2.67 miles) to Overlook Restrooms (L08) } Again, the menu is re-displayed: Menu: 1. View distances to all reachable landmarks. 2. View a report of the proposed landmarks at which first aid stations should be located. 3. Exit TrailManager. Enter your answer:
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Test #5 testID: testViewDistancesInvalidLandmark Strategy: (Exception/unexpected input – invalid landmark ID) Author: Jayani Sivakumar	Preconditions: - TrailManagerUI has been loaded successfully. - The user has loaded landmarkInfo.txt and trails.txt successfully. Steps: - Enter the number 1 in the section “Enter your answer:” that represents “View distances to all reachable landmarks” and press the Enter key. - Enter “L56” as the starting landmark ID and press the Enter key. Check Results	“The provided landmark ID (L56) is invalid for the park.” Again, the menu is re-displayed: Menu: - View distances to all reachable landmarks. - View a report of the proposed landmarks at which first aid stations should be located. - Exit TrailManager. Enter your answer:	Output: Selecting 1 then inputting L56 → Enter produces: “The provided landmark ID (L56) is invalid for the park.” Again, the menu is re-displayed: Menu: - View distances to all reachable landmarks. - View a report of the proposed landmarks at which first aid stations should be located. - Exit TrailManager. Enter your answer:
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